



ICMC 2023-2024

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by oVlad

Round 1 26 November 2023

- 1 Let F_n denote the n -th Fibonacci number. Prove that 3^{2023} divides

$$3^2 \cdot F_4 + 3^3 \cdot F_6 + 3^4 \cdot F_8 + \cdots + 3^{2023} F_{4046}.$$

Proposed by Dylan Toh

- 2 Fredy starts at the origin of the Euclidean plane. Each minute, Fredy may jump a positive integer distance to another lattice point, provided the jump is not parallel to either axis. Can Fredy reach any given lattice point in 2023 jumps or less?

Proposed by Tony Wang

- 3 There are 105 users on the social media platform Mathsenger, every pair of which has a direct messaging channel. Prove that each messaging channel may be assigned one of 100 encryption keys, such that no 4 users have the 6 pairwise channels between them all being assigned the same encryption key.

Proposed by Fredy Yip

- 4 Points A, B, C , and D lie on the surface of a sphere with diameter 1. Determine the maximum possible volume of tetrahedron $ABCD$.

Proposed by Fredy Yip

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-Is there a non-linear integer-coefficient polynomial $P(x)$ and an integer N such that all integers greater than N may be written as the greatest common divisor of $P(a)$ and $P(b)$ for positive integers $a > b$?

-Is there a non-linear integer-coefficient polynomial $Q(x)$ and an integer M such that all integers greater than M may be written as $Q(a) - Q(b)$ for positive integers $a > b$?

Proposed by Dylan Toh

- 6 Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection of the positive integers. Prove that at least one of the following limits is true:

$$\lim_{N \rightarrow \infty} \sum_{n=1}^N \frac{1}{n + f(n)} = \infty; \quad \lim_{N \rightarrow \infty} \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{f(n)} \right) = \infty.$$

Proposed by Dylan Toh

Round 2 25 February 2024

- 1** Prove that there exist distinct positive integers $a_1, a_2, \dots, a_{2024}$ such that for each $i \in \{1, 2, \dots, 2024\}$

$$a_i \mid a_1 a_2 \cdots a_{i-1} a_{i+1} \cdots a_{2024} + k,$$

where a) $k = 1$ and b) $k = 2024$.

Proposed by Ishan Nath

- 2** Let $n \geq 3$ be a positive integer. A circular necklace is called *fun* if it has n black beads and n white beads. A move consists of cutting out a segment of consecutive beads and reattaching it in reverse. Prove that it is possible to change any fun necklace into any other fun necklace using at most $(n - 1)$ moves.

Note: Necklaces related by rotations or reflections are considered to be the same.

Proposed by Dylan Toh

- 3** Let N be a fixed positive integer, S be the set $\{1, 2, \dots, N\}$ and $\mathcal{F}$ be the set of functions $f : S \rightarrow S$ such that $f(i) \geq i$ for all $i \in S$. For each $f \in \mathcal{F}$ let P_f be the unique polynomial of degree less than N satisfying $P_f(i) = f(i)$ for all $i \in S$. If f is chosen uniformly at random from $\mathcal{F}$ determine the expected value of $P'_f(0)$ where

$$P'_f(0) = \left. \frac{dP_f(x)}{dx} \right|_{x=0}.$$

Proposed by Ishan Nath

- 4** Let $(t_n)_{n \geq 1}$ be the sequence defined by $t_1 = 1, t_{2k} = -t_k$ and $t_{2k+1} = t_{k+1}$ for all $k \geq 1$. Consider the series

$$\sum_{n=1}^{\infty} \frac{t_n}{n^{1/2024}}.$$

Prove that this series converges to a positive real number.

Proposed by Dylan Toh

- 5** Is it possible to dissect an equilateral triangle into three congruent polygonal pieces (not necessarily convex), one of which contains the triangle's centre in its interior?

Note: The interior of a polygon is the polygon without its boundary.

Proposed by Dylan Toh
